Avancee en apprentissage statistique

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I missed last week, so this class started in the middle of an uncertainty and evidence theory lecture. There is a difference between a statement being supported by evidence and a statement being compatible with evidence. If I say: "The value is likely somewhere between 3 and 4", then that supports a downstream possibility and presents a domain of claims that are compatible with that previous evidence-supported claim.
We have aleatory uncertainty and epistemic uncertainty:

- **Aleatory Uncertainty:** Uncertainty due to variability in the data-generating process.
- **Epistemic Uncertainty:** Uncertainty due to lack of knowledge about the process.

I have gaps in my knowledge—due to missing the first class—that I must take time to amend. But we are moving on to the next slide set so I will take this opportunity to attempt to follow along with that, and add to this section as I go.
Evidence theory has a basic formal setting. We have a question Q that has one and only one correct answer, which is unknown and we lable as X. We also have Θ, which is the set of all possible answers and is called the frame of discernment. We can get evidence that gives us information about what X *could* be, but it only gives us partial information about it. So the central investigation of this course—I believe it is about evidence theory—is into how that partial information can be represented, propagated, and combined. The separation into aleatory uncertainty—inherent to the process—and epistemic uncertainty—a failure of the modeling—seems to allow us to get more meta about not only what we know but how strong our knowledge is.

1.1 Random Sets

In probability theory, we have random variables. This has an analagous representation in evidence theory, called random sets. It takes the previously finite frame Θ and extends it to work in a concept of infinite frames, with an arbitrary frame Θ and a random closed set in $\mathbb{R}^p$.